

RoadSoS – Technical Documentation

AI-Powered Emergency Response & Roadside Assistance System

1. Project Title

RoadSoS: AI-Powered Emergency Response & Roadside Assistance System

2. Abstract

Road accidents and emergency situations require immediate assistance and rapid communication with emergency responders. Delays in locating victims, contacting emergency services, and informing family members can significantly increase the severity of incidents.

RoadSoS is a web-based emergency response platform designed to provide instant assistance during road emergencies. The system enables users to obtain their current location, identify nearby hospitals and police stations, send SOS alerts via email, manage emergency contacts, and access emergency calling services. The platform leverages modern web technologies and cloud services to provide a scalable, user-friendly, and reliable emergency support solution.

The project aims to reduce emergency response time and improve communication between victims, emergency contacts, and first responders.

3. Introduction

Road accidents are one of the leading causes of injuries and fatalities worldwide. In many cases, victims are unable to communicate their location or contact emergency services quickly.

Existing emergency response mechanisms often require users to manually share location details, which may be difficult during stressful situations. There is a need for a simple, fast, and accessible solution that can automatically provide location information and notify emergency contacts.

RoadSoS addresses this challenge by combining location tracking, emergency communication, authentication, and nearby service discovery into a single platform.

4. Problem Statement

During emergencies, users often face the following challenges:

- Difficulty sharing exact location with responders.
- Delays in contacting hospitals or police stations.

- Lack of centralized emergency information.
- Inability to quickly notify emergency contacts.
- Limited access to emergency assistance while traveling.

These issues can increase response time and negatively affect outcomes during critical situations.

5. Proposed Solution

RoadSoS provides an integrated emergency response platform that enables users to:

- Detect and share their current location.
- Send SOS alerts containing location information.
- Store and manage emergency contacts.
- Locate nearby hospitals and police stations.
- Access emergency calling services.
- Maintain SOS alert history.
- Use voice-assisted emergency features.

The system is accessible through any web browser and can be used on both desktop and mobile devices.

6. Objectives

The primary objectives of RoadSoS are:

1. Provide instant access to emergency services.
 2. Reduce response time during emergencies.
 3. Improve communication between victims and responders.
 4. Enable automatic location sharing.
 5. Provide easy access to nearby hospitals and police stations.
 6. Enhance user safety through emergency notifications.
 7. Create a scalable and cloud-based emergency response platform.
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7. Scope of the Project

The project focuses on emergency assistance for road users and travelers.

Included Features

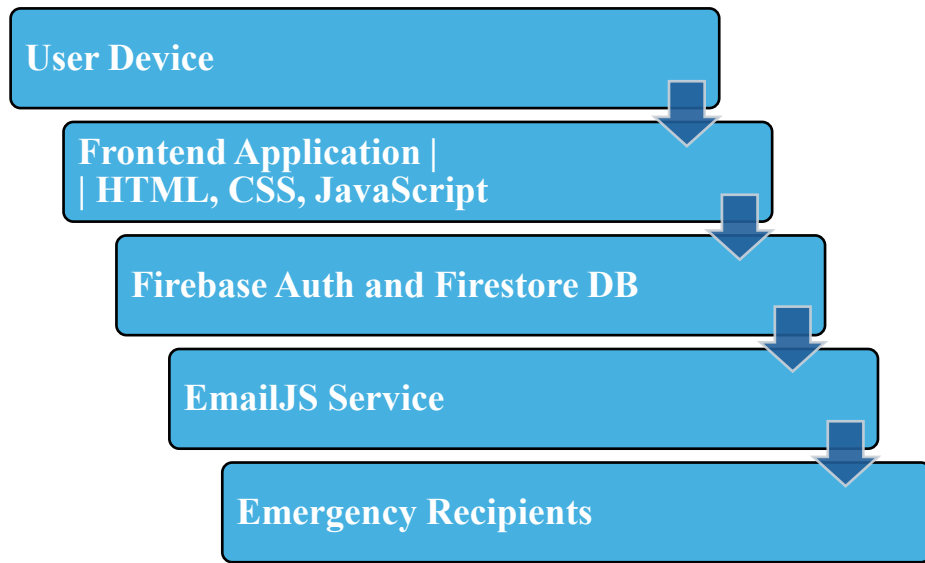
- User Authentication
- GPS Location Tracking
- SOS Email Alerts
- Emergency Contact Management
- Nearby Hospital Finder
- Nearby Police Station Finder
- Voice Assistant
- Emergency Calling
- SOS History Management
- Emergency Report Generation

Future Scope

- Mobile Application
- SMS-Based Emergency Alerts
- Real-Time Ambulance Tracking
- AI-Based Accident Detection
- Push Notifications
- Multi-Language Support
- Admin Monitoring Dashboard

8. System Architecture

Architecture Overview



9. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript (ES6)

Backend Services

- Firebase Authentication
- Firebase Firestore Database
- Firebase Hosting

Third-Party Services

- EmailJS
- Geolocation API
- OpenStreetMap Services

Development Tools

- Visual Studio Code
- Git
- Firebase CLI
- Google Chrome

10. Functional Requirements

User Authentication

- User Registration
- User Login
- User Logout
- Secure Session Management

Location Services

- Obtain Current GPS Coordinates
- Display Latitude and Longitude
- Open Location in Google Maps

Emergency Contacts

- Add Emergency Contacts
- Store Contacts Securely
- User-Specific Contact Management

SOS Alert System

- Generate Emergency Alert
- Share User Location
- Send Email Notifications
- Maintain Alert History

Emergency Services

- Access Emergency Calling Numbers
- Find Nearby Hospitals
- Find Nearby Police Stations

11. Non-Functional Requirements

Performance

- Fast response time
- Real-time location retrieval

Reliability

- Cloud-hosted architecture
- Continuous availability

Scalability

- Supports increasing users
- Expandable database structure

Security

- Firebase Authentication
- Protected user data
- Secure cloud storage

Usability

- Simple user interface
- Mobile-friendly design
- Easy navigation

12. Database Design

Firestore Collections

Contacts Collection

```
{  
  "name": "Father",  
  "email": "father@gmail.com",  
  "owner": "user@gmail.com"  
}
```

SOS History Collection

```
{  
  "location": "https://maps.google.com/?q=17.67,75.95",  
  "createdAt": "2026-06-22 10:30"  
}
```

13. Key Features

1. User Authentication

Allows users to securely create accounts and access personalized emergency data.

Benefits

- Personalized experience
 - Secure data management
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2. GPS Location Tracking

Retrieves user location using the browser Geolocation API.

Benefits

- Accurate location sharing
 - Faster emergency response
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3. SOS Alert System

Sends emergency alerts with location information using EmailJS.

Benefits

- Immediate communication
 - Quick assistance
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4. Nearby Hospital Finder

Identifies nearby hospitals based on user location.

Benefits

- Faster medical access
-

5. Nearby Police Station Finder

Displays nearby police stations.

Benefits

- Enhanced safety support
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6. Emergency Contact Management

Allows users to store emergency contacts linked to their account.

Benefits

- Personalized emergency communication
-

7. Voice Assistant

Provides voice-based interaction and accessibility.

Benefits

- Hands-free operation
-

8. Emergency Calling

Provides one-click emergency calling options.

Benefits

- Quick emergency access
-

14. Implementation Methodology

Phase 1

Requirement Analysis

Phase 2

System Design

Phase 3

Frontend Development

Phase 4

Firebase Integration

Phase 5

SOS Email Integration

Phase 6

Testing and Debugging

Phase 7

Deployment on Firebase Hosting

15. Testing

Functional Testing

Test Case	Expected Result	Status
User Signup	Account Created	Pass
User Login	Login Successful	Pass
Logout	Session Ended	Pass
Get Location	Location Displayed	Pass
SOS Alert	Email Sent	Pass
Add Contact	Contact Saved	Pass
Hospital Finder	Results Displayed	Pass
Police Finder	Results Displayed	Pass

16. Results

The developed system successfully:

- Authenticates users.
- Retrieves real-time location.
- Sends emergency alerts.
- Stores emergency contacts.
- Displays nearby hospitals and police stations.
- Maintains SOS history.
- Provides emergency communication tools.

The platform demonstrates the feasibility of a cloud-based emergency response solution for road safety applications.

17. Future Enhancements

- AI-Based Accident Detection
- SMS Alert Integration
- Mobile Application Development
- Push Notifications

- Real-Time Ambulance Tracking
 - Advanced Analytics Dashboard
 - Multi-Language Support
 - Emergency Responder Integration
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18. Conclusion

RoadSoS is a comprehensive emergency response and roadside assistance platform that improves safety by enabling rapid communication, location sharing, and emergency service access. The solution successfully demonstrates how modern web technologies and cloud services can be integrated to provide practical and scalable emergency assistance.

The project achieves its objective of reducing emergency response delays while improving accessibility and user safety. With additional enhancements, RoadSoS can evolve into a large-scale emergency management ecosystem capable of serving a wider population and supporting advanced emergency response operations.

19. References

1. Firebase Documentation – <https://firebase.google.com/docs>
 2. EmailJS Documentation – <https://www.emailjs.com/docs>
 3. MDN Web Docs – <https://developer.mozilla.org>
 4. OpenStreetMap – <https://www.openstreetmap.org>
 5. Google Maps – <https://maps.google.com>
 6. W3Schools – <https://www.w3schools.com>
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Project Name: RoadSoS

Version: 1.0

Deployment URL: <https://roadsos-b7e23.web.app>

Technology: HTML, CSS, JavaScript, Firebase, EmailJS

Status: Functional MVP Ready for Evaluation 🚀 🏆